

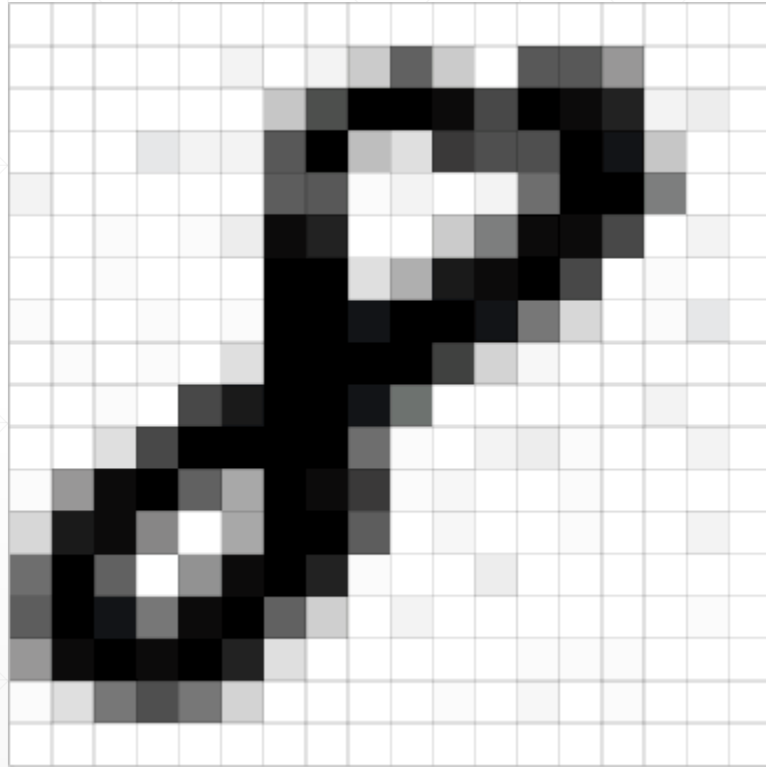
Hand-written Digits Recognition

- MNIST
 - 7000 images per category
 - train/test splitting: 60k vs 10k



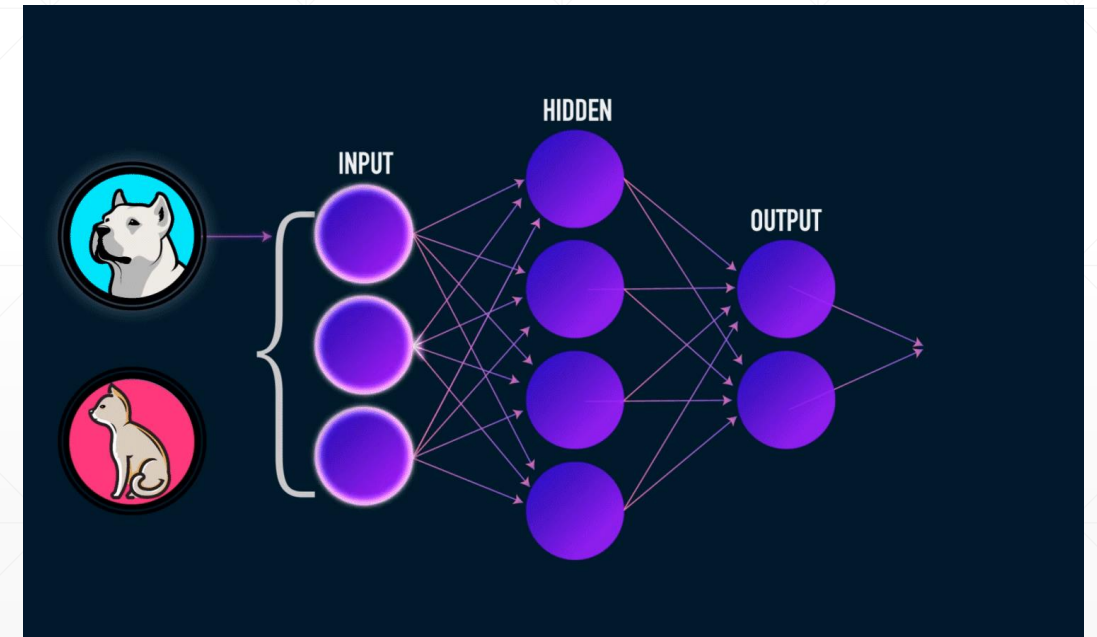
Image

- [28,28,1]
- → [784]



Input and Output

- $x: [b, 784]$
- prediction:
 - 1. dog = 0, cat = 1, fish = 2, ...
 - 2. dog = [1, 0, 0, ...]
 - cat = [0, 1, 0, ...]
 - fish = [0, 0, 1, ...]



Regression VS Classification

- $y = w * x + b$
 - $y \in R^d$
 - $out = X@W + b$
 - out: [0.1, 0.8, 0.02, 0.08]
 - $pred = \operatorname{argmax}(out)$
 - pred: 1
 - label: 2
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Computation Graph

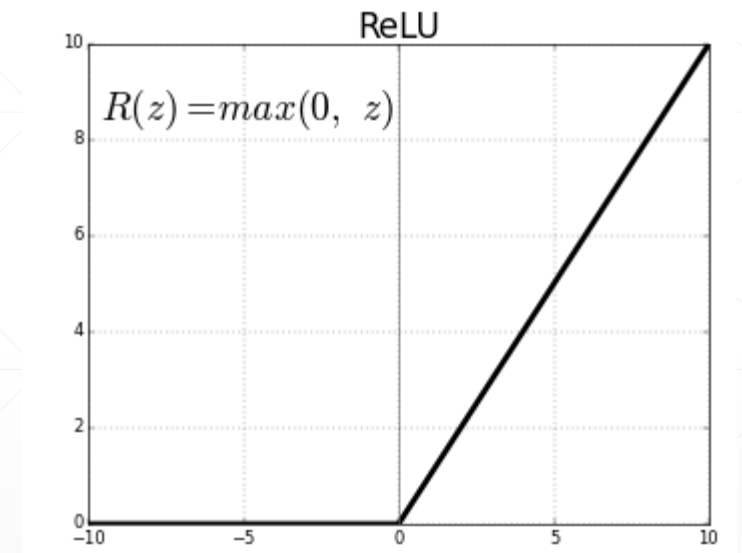
- $out = X@W + b$
 - X: [b, 784]
 - W: [784, 10]
 - b: [10]
 - out: [b, 10]
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It's Linear!

- $out = X@W + b$

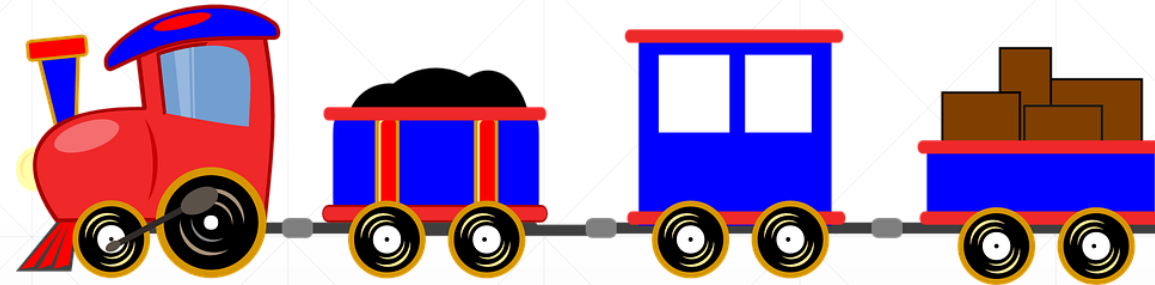
- $out = f(X@W + b)$

- $out = \text{relu}(X@W + b)$



It's too simple!

- $out = \text{relu}(X@W + b)$
- $h_1 = \text{relu}(X@W_1 + b_1)$
- $h_2 = \text{relu}(h_1@W_2 + b_2)$
- $out = \text{relu}(h_2@W_3 + b_3)$



Particularly

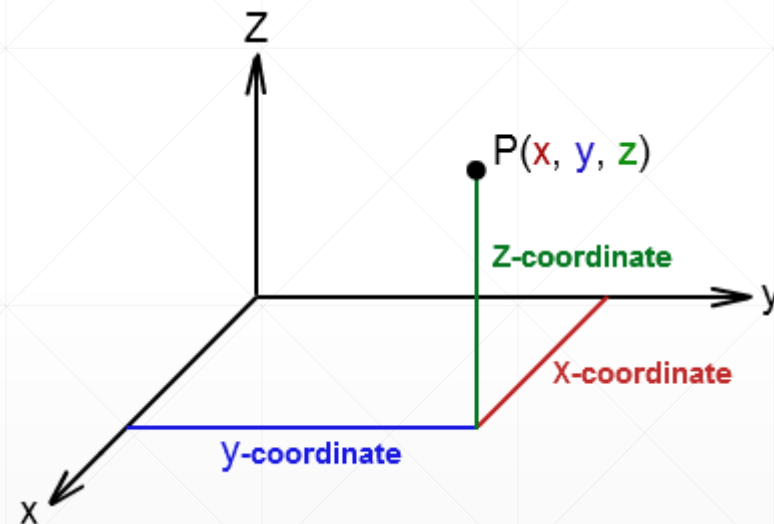
- $X = [v_1, v_2, \dots, v_{784}]$
 - $X: [1, 784]$
- $h_1 = \text{relu}(X @ W_1 + b_1)$
 - $W_1: [784, 512]$
 - $b_1: [1, 512]$
- $h_2 = \text{relu}(h_1 @ W_2 + b_2)$
 - $W_2: [512, 256]$
 - $b_2: [256]$
- $out = \text{relu}(h_2 @ W_3 + b_3)$
 - $W_3: [256, 10]$
 - $b_3: [10]$



[0, 0, 0.01, 0.1, 0.8, 0, ...]

Loss?

- *out*: [1, 10]
- Y/label: 0~9
 - eg.: 1 → [0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0]
 - eg.: 3 → [0, 0, 0, 3, 0, 0, 0, 0, 0, 0, 0]
- Euclidean Distance: *out* → Label
 - MSE



In a nutshell

- $out = \text{relu}\{\text{relu}\{\text{relu}[X@W_1 + b_1]@W_2 + b_2\}@W_3 + b_3\}$
 - $pred = \text{argmax}(out)$
 - $loss = \text{MSE}(out, label)$
 - minimize $loss$
 - $[W'_1, b'_1, W'_2, b'_2, W'_3, b'_3]$
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Deep Learning?

- We have not seen it.
- But we already master it.
- We will show you It's (almost) Deep Learning!



Classification Procedure

- Step1. Compute $[h_1, h_2, out]$
 - Step2. Compute $Loss$
 - Step3. Compute gradient and update $[W'_1, b'_1, W'_2, b'_2, W'_3, b'_3]$
 - Step4. Loop
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